MULTIMACSIM.v1 — B.Tech Year 1 Electrical Engineering Edition

Scope & Target Curriculum

`MULTIMACSIM.v1` is specifically tailored to cover 100% of the **B.Tech 1st Year Basic Electrical Engineering (BEE) & Network Analysis** syllabus:

Core Theory & Lab Coverage

1. **Fundamental Network Laws:** Ohm's Law, Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL).
2. **Network Theorems:**
   - **Thevenin's Theorem:** Finding open-circuit voltage V_{th} Vth and Thevenin resistance R_{th} Rth.
   - **Norton's Theorem:** Finding short-circuit current I_N IN and Norton resistance R_N RN.
   - **Maximum Power Transfer Theorem:** Verifying that maximum power is transferred when $R_L = R_{th}$ RL = Rth, plotting P_{RL} PRL vs R_L RL.
   - **Superposition Theorem:** Enabling/disabling individual independent sources.
3. **Controlled / Dependent Sources:**
   - Voltage-Controlled Voltage Source (VCVS): $V = \mu \cdot V_c$ V = μ · Vc
   - Voltage-Controlled Current Source (VCCS): $I = g_m \cdot V_c$ I = gm · Vc
   - Diamond schematic symbols with control terminals.
4. **AC Sinusoidal Steady-State & Phasor Analysis:**
   - Phasor representation ($\vec{V}$ V⃗ and $\vec{I}$ I⃗).
   - Phase difference angle $\phi = \theta_v - \theta_i$ ϕ = θv − θi.
   - **Power Triangle:**
     - Active / Real Power: $P = V_{\text{RMS}} I_{\text{RMS}} \cos\phi$ P = VRMSIRMS cos ϕ (Watts)
     - Reactive Power: $Q = V_{\text{RMS}} I_{\text{RMS}} \sin\phi$ Q = VRMSIRMS sin ϕ (VAR)
     - Apparent Power: $S = V_{\text{RMS}} I_{\text{RMS}}$ S = VRMSIRMS (VA)
     - Power Factor: $\text{PF} = \cos\phi$ PF = cos ϕ (Lagging / Leading indicator)

5. **Series & Parallel RLC Resonance:**
    - Resonant frequency $f_0 = \frac{1}{2\pi\sqrt{LC}}$f0 = 2π LC

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    - Quality factor $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$Q = Rω0L = ω0CR1
    - Bandwidth $\mathrm{BW} = \frac{f_0}{Q}$BW = Qf0
6. **Diode Applications:**
    - Half-Wave Rectifier
    - Full-Wave Bridge Rectifier (4-diode bridge with smoothing filter capacitor)

Pre-Loaded 1st-Year Academic Experiments in v1 (Completed & Verified)

- ✓
  Lab Exp 1: KVL & KCL Loop & Nodal Verification
- ✓
  Lab Exp 2: Thevenin's & Norton's Theorem Equivalence
- ✓
  Lab Exp 3: Maximum Power Transfer Curve (R_L RL sweep)
- ✓
  Lab Exp 4: Series RLC Resonance & Impedance Minimum ($f_0 = 1591.5\ \mathrm{Hz}$f0 = 1591.5 Hz)
- ✓
  Lab Exp 5: Full-Wave Bridge Rectifier with Filter & Ripple

Verification & Status

- **Test Suite Status:** 9/9 Tests Passing (100% Pass Rate in 0.011s)
- **App Status:** Successfully compiled and ready to run with `swift run` or in Xcode.